Angela Magtoto

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Education

University of California, Los Angeles — **June 2028**
Bachelor's of Science, Electrical Engineering — *Los Angeles, CA*
Coursework: Systems & Signals, Computer Vision, Circuit Theory, Digital Logic Design, Data Structures & Algorithms, Assembly
Activities: AWS Cloud Club, Theta Tau, Bruin Health Consulting, DataRes, IEEE, Pilipinos in Engineering, Pilipino Cultural Night

Work Experience

Amazon Web Services (AWS) — **June 2025 – Sept 2025**
Software Development Engineering Intern — AI/ML Science Team — *New York, NY*
- Developed a multi-agent system using Agentic protocols (A2A) and AWS Cloud, reducing manual customer feedback analysis, and saving hours weekly spent on repetitive document review
- Addressed security concerns by building a custom MCP server with OAuth per-user isolation, ensuring secure API integration using FastMCP SDK and AWS Bedrock AgentCore
- Tackled scalability issues by developing a Lambda-triggered RAG pipeline, delivering faster insight extraction from previously inaccessible unstructured data
- Developed a TypeScript frontend with streaming response handling and agent memory state management
- Authored an article on multi-agent system design, adopted as a reference for agentic AI onboarding across my organization

UCLA Mobility Lab — **Sept 2025 – Present**
Undergraduate Researcher — Autonomous Driving & Computer Vision — *Los Angeles, CA*
- Designed an NLP pipeline that translates natural language driving descriptions into executable CARLA scenarios by mapping RAG-extracted JSON constraints to XML route files, integrating a custom planner that validates difficulty
- Developed an extensive benchmarking suite to profile single vs. multi-agent autonomous driving models, using pre-crash scenarios to quantify safety metrics including collision avoidance rate and robustness
- Ran and analyzed CARLA simulations of autonomous driving to generate a proprietary dataset of behaviors for analyzing model performance in 120+ cooperative driving scenarios

Wedge (Y-Combinator S25) — **Jan 2026 – April 2026**
Product Management Intern — *Remote*
- Engineered a Chrome extension in JavaScript automating personalized Gmail draft generation for marketing outreach, using the OpenAI API to generate personalized email copy for each contact
- Built a CSV ingestion pipeline to parse Apollo contact exports & populate email templates with company specific variables

Projects

Micromouse | IEEE @ UCLA — **C++ | KiCAD**
- Collaborated with a team to implement C++ pathfinding algorithms, achieving a sub-30 second maze completion
- Designed a custom PCB to integrate a microcontroller with all peripheral navigation sensors and motor control systems

AWS Autonomous Car (Deepracer) — **Python | PyTorch | Reinforcement Learning**
- Trained a CNN in PyTorch for real-time visual track recognition and steering prediction on an embedded racing platform
- Optimized reward function using waypoint-based incentives and heading penalties, improving lap consistency by 75%

Combat Robot — **Python | SolidWorks**
- Developed a 3lb combat robot using SolidWorks, performing FEA to optimize structural integrity of the moving system
- Programmed adaptive PID controller to manage stability during high-RPM weapon spinning, optimizing maneuverability

Skills

Languages: Python (NumPy, SciPy, Pandas), C++, TypeScript, JavaScript, SQL, Swift
AI/Cloud: AWS (Lambda, S3, Bedrock, Kendra, AgentCore), Docker Deployment, FastMCP, PyTorch, CARLA
Hardware: PCBA Design, KiCAD, Altium, SolidWorks, MATLAB, Git

Leadership

Hackathon Organizer | *Spark & Hack the Wave* — **2022 – 2025**
- Hosted two hackathons to expand gender diversity and accessibility in technology, featured in Forbes article
- Organized workshops for 90+ students, mentoring participants on NLP fundamentals and full-stack web development

Campus Ambassador | Amazon, Adobe, Congressional App Challenge